

# TrueLearn OPP3 Quiz

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# Question 1

A 1-year-old male presents to the office with his mother, who reports that he has had a fever and irritability. He has also been pulling at his right ear for the past 2 days. Vital signs reveal a temperature of 38.9°C (102.1°F). Physical examination reveals a bulging, erythematous tympanic membrane with a decreased cone of light. Osteopathic manipulative therapy\* should be directed toward which of the following bones?

\*Please note that this question is copied directly from the Qbank, but please note that you practice Osteopathic Manipulative TREATMENT (OMT) not Therapy (OMTh). OMTh is what is practiced by international osteopaths who are not fully trained and licensed physicians! ☺

#### Explanation

The Eustachian tube is closely related to the **temporal bone**. The Eustachian tube travels through the petrous portion of the temporal bone itself. As it exits the temporal bone, it adheres to the cranial base in a cartilaginous sheath. The sheath is contained between the greater wing of the sphenoid and the petrous portion of the temporal bone. It emerges into the posterior pharynx. Dysfunctions of the temporal and/or sphenoid bones will cause strain and improper orientation of the Eustachian tube. The orientation of the Eustachian tube changes with the maturation of the cranial base as the child ages and is generally in the adult position by 6 years of age.

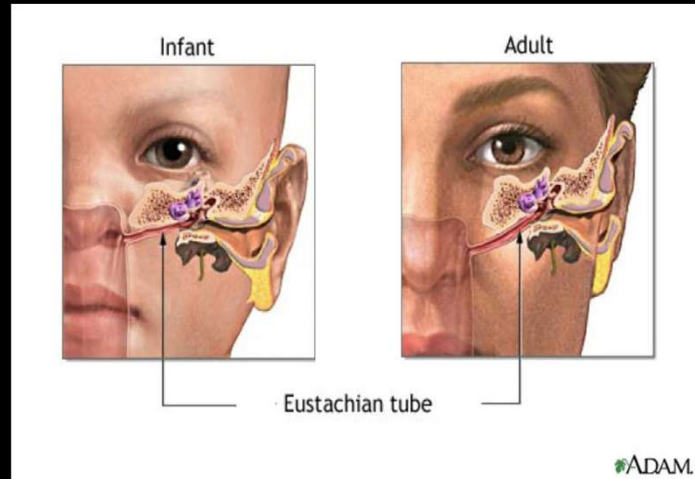
Treatment of the Eustachian tube can be accomplished with cranial manipulation of the temporal and/or sphenoid bones. This allows for proper orientation of the temporal bone and appropriate drainage to occur within the Eustachian tube(s).

## Explanation and Bottom Line

Bottom Line: pts dx with AOM (acute otitis media) are likely to suffer from temporal bone dysfunction on the affected side.

## Eustachian tube

- Kids vs adults
  - More pliable and easily compressible
  - More horizontally oriented
  - Shorter
  - Poor ability to equilibrate pressure
- Changes after ~7 y/o
- Proximal 1/3 is bony, remainder is fibrocartilaginous
- Muscles: tensor veli palatini, levator veli palatini, salpingopharyngeus, tensor tympani



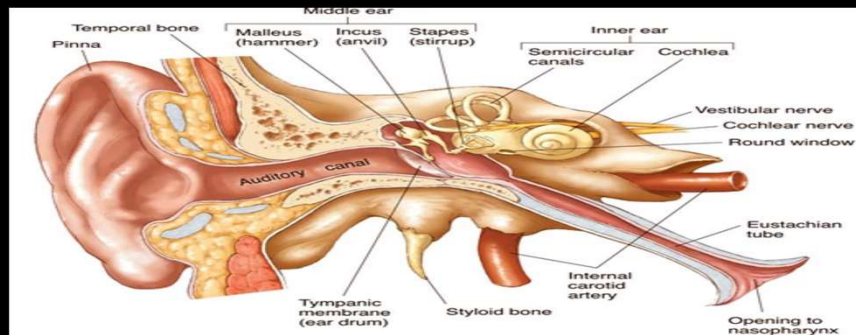
Adult angle is 45 degrees and kids is 10 degrees

Also feed supine, reflux, etc

10 mm infants, 36 mm adults

# Osseus Considerations for AOM

- The middle/inner ear and bony portion Eustachian tubes lie within temporal bone
  - Squamous and petrous portions
- Distal Eustachian tubes pass through groove between petrous portion temporal bone and greater wing of sphenoid



## Distractors

**Answer A:** Treatment of the frontal bone would be an appropriate treatment in case of an upper respiratory infection or sinusitis and these are common causes of otitis media, but treating the frontal bone would not directly influence the Eustachian tube. Therefore, it is not the best answer. Treating restricted frontal bones can help to drain the frontal, ethmoid, and sphenoid sinuses and can help with dacryostenosis because of the frontal's influence on the orbit and, specifically, the nasal bones.

**Answer B:** Treatment of the maxilla would not directly influence the Eustachian tube, although treatment of the maxilla would be effective in treating maxillary sinusitis, dental problems, and upper respiratory infections.

**Answer C:** Treatment of the occiput would indirectly influence the Eustachian tube because of the occiput's direct articulations with the temporal bone. This is not the best answer, however. Treatment of the occiput is very useful in infants with suckling difficulties, colic, or gastroesophageal reflux.

**Answer D:** Treatment directed towards the parietal bones would indirectly influence the temporal bones because of the parietal bones' direct articulations with the temporal bones. This is not the best answer, however. Treatment of the parietal bones helps patients, especially infants, with molding and plagiocephaly. Treatment of this bone can also benefit adults and children with headaches.

## Question 2

A 45-year-old male presents to the ED with upper abdominal pain, nausea, and vomiting. The pain is less severe when he bends forward at the waist. PMH reveals alcohol use disorder. PE reveals severe tenderness in the epigastrium and ecchymotic discoloration in the flanks. Given his most serious acute problem, a Chapman point will most likely be located at which location?

# Explanation

- Bottom Line: The anterior Chapman point for the pancreas is at 7<sup>th</sup> intercostal space on the right and the posterior point is at T7 on the right, midway between the tip of the right TP and the SP.

This patient's most serious acute problem is likely **acute necrotic pancreatitis**, as suggested by his epigastric pain relieved by bending forward ("pancreatic position"), nausea, vomiting, and ecchymotic discoloration of the flank ("Grey Turner sign"). The pancreatic position reflects pain improvement as pressure on the parietal peritoneum is reduced when bending forward. The Grey Turner sign reflects necrosis of the pancreas with hemorrhage. The patient's history of alcohol use disorder further supports the diagnosis, as the 2 most common causes of pancreatitis are gallstones and alcoholism.

**Chapman points** (also called Chapman reflex points) represent viscerosomatic reflexes. There are both anterior and posterior Chapman points associated with each internal organ. The anterior Chapman point for the pancreas is located in the 7<sup>th</sup> intercostal space on the right, while the posterior Chapman point is located at **T7 on the right**, midway between the tip of the right transverse process and the spinous process.

The pancreas receives its sympathetic innervation primarily from T5 to T9 via the greater splanchnic, but it also receives some innervation from T10 to T11 via the lesser splanchnic nerve. Sympathetic viscerosomatic reflex changes may therefore be found at T5 to T11. The pancreas receives its parasympathetic innervation from the vagus nerve and may therefore present with parasympathetic viscerosomatic changes in the upper cervical (OA and AA) and cranial areas.

## Distractors

**Answer B:** T11 on the right would be the location of the posterior Chapman point for the appendix. While appendicitis would lead to nausea and vomiting, it would present with lower abdominal, not epigastric, pain. Furthermore, appendicitis would not explain the patient's pancreatic position and Grey Turner sign.

**Answer C:** The 5th intercostal space on the right corresponds to the anterior Chapman point for the liver. While a patient with alcohol use disorder may also have a somatic dysfunction in this location, the patient's most serious problem at present is acute pancreatitis. The anterior Chapman point for the pancreas is located in the 7th intercostal space on the right.

**Answer D:** The 6th intercostal space on the left is an anterior Chapman point for the stomach (related to peristalsis). While this patient who has been vomiting may have somatic dysfunction at this location, vomiting itself is not his most serious issue. Acute pancreatitis is his most important current problem, which would present with an anterior Chapman point in the 7th intercostal space on the right and a posterior Chapman point at T7 on the right.

**Answer E:** The 6th intercostal space on the right refers to the anterior Chapman point for the liver and gallbladder. While this patient with alcohol use disorder may have a somatic dysfunction in this location because of chronic liver disease, his most serious acute problem is acute pancreatitis.

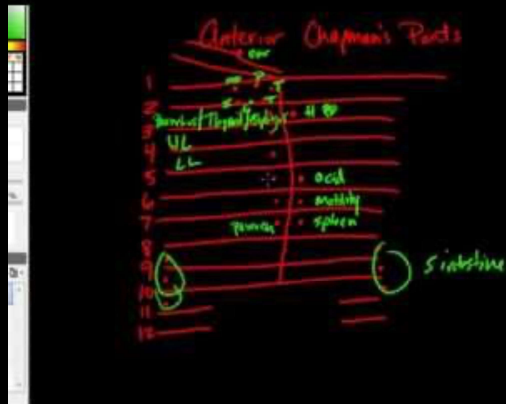
# GI Chapman's Points

| Gastrointestinal Chapman Points   |                                               |                                                           |
|-----------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| Organ/Structure                   | Anterior                                      | Posterior*                                                |
| Esophagus                         | 2nd ICS                                       | T2–T3 bilaterally                                         |
| Stomach (acidity)                 | 5th ICS on the left                           | T5 on the left                                            |
| Stomach (peristalsis)             | 6th ICS on the left                           | T6 on the left                                            |
| Liver                             | 5th–6th ICS on the right                      | T5–T6 on the right                                        |
| Gallbladder                       | 6th ICS on the right                          | T6 on the right                                           |
| Pancreas                          | 7th ICS on the right                          | T7 on the right                                           |
| Spleen                            | 7th ICS on the left                           | T7 on the left                                            |
| Small bowel                       | 8th, 9th, 10th ICS bilaterally                | T8–T10 bilaterally                                        |
| Appendix                          | Tip of 12th rib on the right                  | T11–T12 on the right                                      |
| Cecum                             | Upper 1/5 of the right anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Ascending colon                   | Middle 3/5 of the right anterior thigh        | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Right 1/2 of the transverse colon | Lower 1/5 of the right anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Left 1/2 of the transverse colon  | Lower 1/5 of the left anterior thigh          | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Descending colon                  | Middle 3/5 of the left anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Rectum                            | Proximal inner thigh on the lesser trochanter | Sacrum at the level of S2                                 |

ICS = intercostal space; TP = transverse process.  
 \*All points from the esophagus to the appendix are located midway between the tip of the TP and the spinous process.

\*\*\*Pearl: Chapman Points are VERY HIGH YIELD since they are easily testable. Study these tables and know them inside and out! They commonly love to give an anterior point and ask where the posterior one is for the same organ or vice versa. Some of the highest yield on this table are the appendix, pancreas, stomach (acidity – like in a GERD case), and whole colon (remember it can be mapped onto your thighs if you imagine reflecting the colon down).

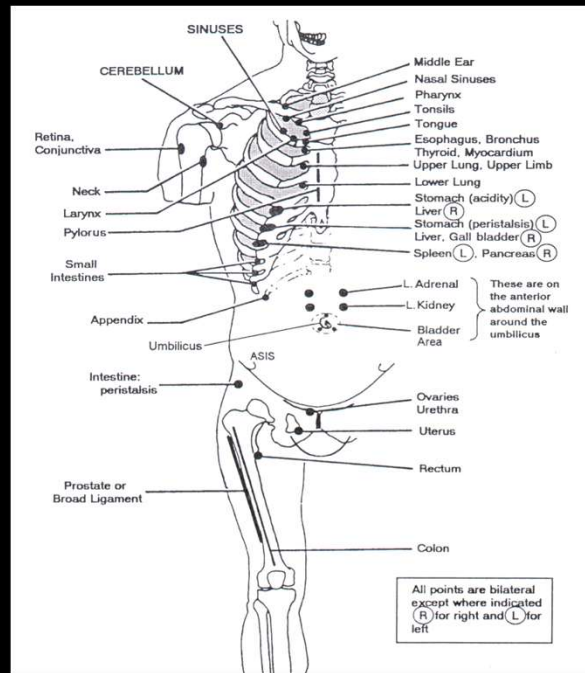
## Anterior Chapman's Points Mnemonic Video



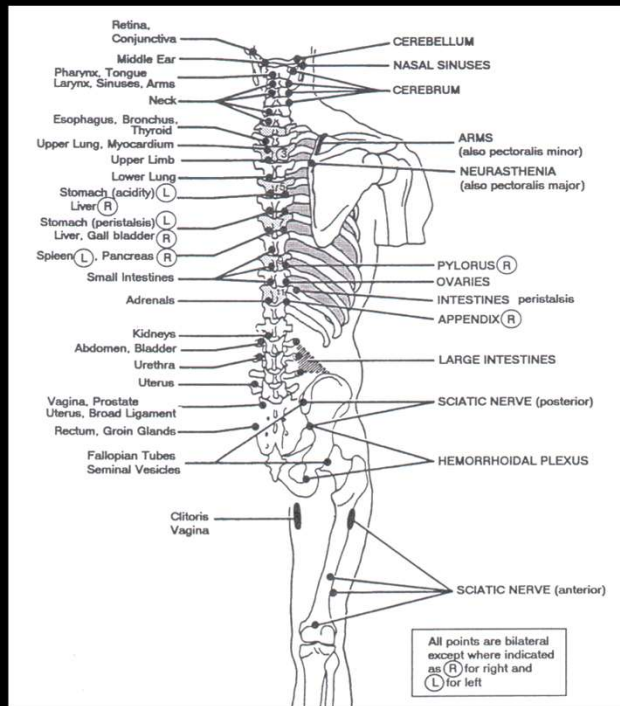
\*\*\*Recommended by past students and OMM Fellows!

## Chapman's Points

- Neurolymphatic reflex
- Anterior and posterior points to screen and treat

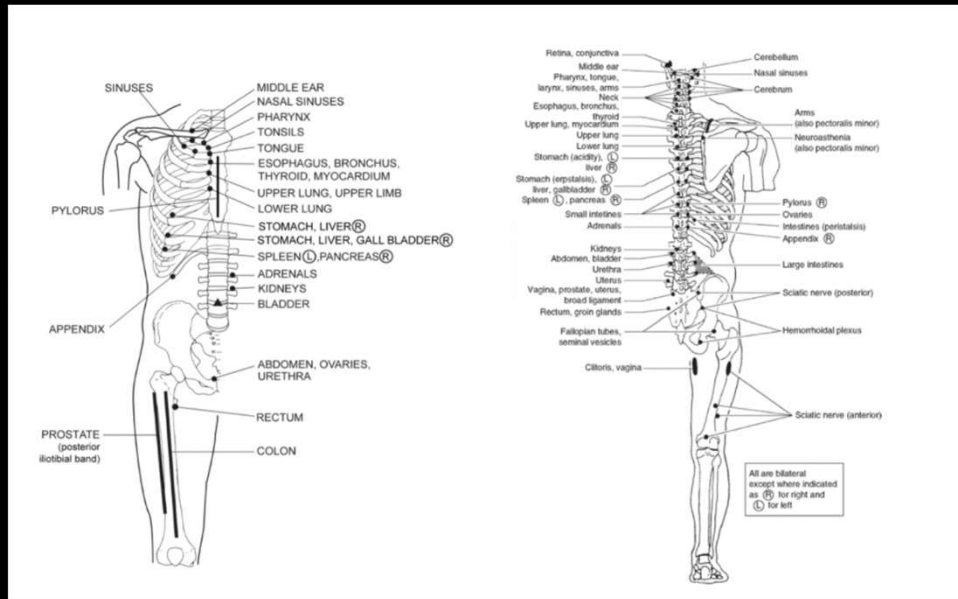


Osteopathic Considerations in Systemic Disease, 2nd Ed pp 232-233

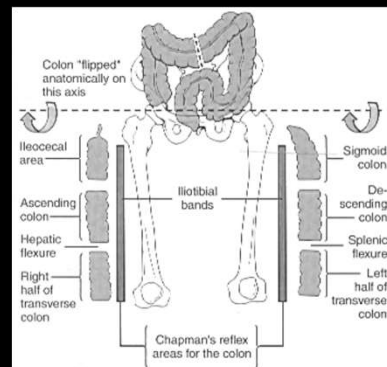


Osteopathic Considerations in Systemic Disease, 2nd Ed pp 232-233

## Chapman's Points



| Location                                                                                                           | Anterior                                                                                                                                                                                                                                                                                          | Posterior                                                                                   |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Upper respiratory tract                                                                                            | Near clavicle, rib 1 or 2                                                                                                                                                                                                                                                                         | C1 or C2                                                                                    |
| Lower respiratory tract<br>Bronchi<br>Upper lung<br>Lower lung                                                     | Rib 2 & 3 at sternocostal junction<br>Rib 3 & 4 at sternocostal junction<br>Rib 4 & 5 at sternocostal junction                                                                                                                                                                                    | T2<br>T3/4<br>T4/5                                                                          |
| Upper GI tract<br>Esophagus<br>Stomach<br>Liver, gallbladder<br>Pancreas<br>Spleen<br>Duodenum<br>Jejunum<br>Ilium | Rib 2 & 3 at sternocostal junction<br>Ribs 5-7 left<br>Ribs 5-7 right<br>Ribs 7 & 8 at costochondral junction right<br>Ribs 7 & 8 at costochondral junction left<br>Ribs 8 & 9 near costochondral junction<br>Ribs 9 & 10 near costochondral junction<br>Ribs 10 & 11 near costochondral junction | T2<br>T5-7 (left)<br>T5-7 (right)<br>T7-8 (right)<br>T7-8 (left)<br>T8,9<br>T9,10<br>T10,11 |
| Lower GI tract<br>Appendix<br>Colon                                                                                | Tip of rib 12 right<br>See figure below                                                                                                                                                                                                                                                           | TP's of T11/T12<br>L2-L4 to iliac crest                                                     |
| Heart                                                                                                              | Between ribs 2&3 at sternocostal junction                                                                                                                                                                                                                                                         | T2/3                                                                                        |
| Urinary system<br>Kidney<br>Bladder<br>Urethra                                                                     | 1" above umbilicus either side of midline<br>Periumbilical and pubic symphysis<br>Pubic symphysis                                                                                                                                                                                                 | T12, L1<br>TP of L2<br>TP of L2                                                             |
| Reproductive organs<br>Ovaries/testes<br>Uterus<br>Prostate                                                        | Anterior pubic bone<br>Medial border of obturator foramen<br>Lateral femur (similar to colon)                                                                                                                                                                                                     | T9-T11<br>TP of L5<br>TP of L5 to PSIS                                                      |



In general anterior points are located at the sternocostal or costochondral junction unless otherwise specified in the table above, and posterior points are located between the spinous process and transverse process of the vertebrae. For example, if the table above states the posterior point is T2, that implies the Chapman's point is midway between the spinous process and transverse process of T2. Likewise T3/T4 implies between the transverse process of T3 and T4, midway between the transverse and spinous process.

Savarese OMT Review,  
[http://files.academyofosteopathy.org/convo/2016/Handouts/Devine\\_SAAOLecture.pdf](http://files.academyofosteopathy.org/convo/2016/Handouts/Devine_SAAOLecture.pdf)

# Sympathetic Innervation of the GI System

| Sympathetic Innervation of the Gastrointestinal system |                     |              |                   |                                                                                                                                                                                                          |
|--------------------------------------------------------|---------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nerve                                                  | Ganglion            | Spinal Level | Area              | Organs                                                                                                                                                                                                   |
| Greater splanchnic                                     | Celiac              | T5–T9        | Upper GIT/foregut | <ul style="list-style-type: none"> <li>• Stomach</li> <li>• Liver</li> <li>• Gallbladder</li> <li>• Spleen</li> <li>• Portions of the pancreas and duodenum</li> </ul>                                   |
| Lesser splanchnic                                      | Superior mesenteric | T10–T11      | Middle GIT/midgut | <ul style="list-style-type: none"> <li>• Portions of the pancreas and duodenum</li> <li>• Jejunum</li> <li>• Ileum</li> <li>• Ascending colon</li> <li>• Proximal 2/3 of the transverse colon</li> </ul> |
| Least splanchnic                                       | Inferior mesenteric | T12–L2       | Lower GIT/hindgut | <ul style="list-style-type: none"> <li>• Distal 1/3 of the transverse colon</li> <li>• Descending colon</li> <li>• Sigmoid colon</li> <li>• Rectum</li> </ul>                                            |
| GIT = gastrointestinal tract.                          |                     |              |                   |                                                                                                                                                                                                          |

\*\*\*Pearl: they love to overlap testing knowledge of Chapman points and viscerosomatic levels like this table shows, so this is another very high-yield table.

## Question 3

33 y/o F presents to the office with low back pain. The pain began when she stood up after bending over for hours while working in her garden earlier that day. Physical examination reveals reduced active lumbar extension, as well as tenderness and warmth in the paraspinal tissues of L2-L5. The most likely additional finding is:

## Explanation and Bottom Line

**Bottom Line:** Acute SD may present with swelling, warmth, moisture, fullness, boggiess, hypertonic muscles, edema, tension, erythema, and tenderness.

This patient likely has **acute somatic dysfunction**, as supported by the injury occurring within the previous 24 hours and the presence of tenderness and warmth in the area of injury. Of the answer choices, **moist skin** is the only additional finding consistent with acute somatic dysfunction. In general, injuries within the previous 1 week are considered acute; otherwise, they are considered chronic.

In acute somatic dysfunction, sympathetic activity is increased systemically, but bradykinin is released in the area of injury, producing local vasodilation. Muscle tone increases via activation of muscle spindle reflexes. Blood vessels are injured, resulting in the release of endogenous peptides that cause inflammation and edema. Soft tissues become boggy as fluid leaks from the vessels. Therefore, the local skin and soft tissue changes of acute somatic dysfunction include swelling, warmth, moisture, fullness, boggiess, hypertonic muscles, edema, tension, erythema, and tenderness.

An easy way to remember the features of acute somatic dysfunction is to recall the *5 cardinal signs of inflammation*:

1. Dolor (pain)
2. Calor (heat)
3. Rubor (redness)
4. Tumor (swelling)
5. Functio laesa (loss of function)

In chronic somatic dysfunction, local vasoconstriction is present because of sympathetic hyperactivity near the site of injury, but systemic sympathetic tone is only slightly increased or has returned to normal. Muscle tone is decreased, and range of motion is often limited because of fascial contracture. Somatosomatic reflexes are commonly observed. Chronic somatic dysfunction is characterized by cool temperature, dry and thickened skin, prolonged blanching, and dull achy pain in the affected area.

## Distractors

**Answer A:** Cool skin would not be an expected tissue texture change in this patient with acute somatic dysfunction. It is a characteristic of chronic somatic dysfunction. Cool skin is caused by localized vasoconstriction from prolonged increased sympathetic tone.

**Answer B:** Fibrotic soft tissues are a characteristic of chronic, not acute, somatic dysfunction. Soft tissue texture in chronic somatic dysfunction may also be described as congested, stringy, doughy, or thickened. The tissues exhibit increased resistance to palpation. With acute somatic dysfunction, soft tissues feel boggy or edematous because of vascular leakage and the effects of peptides in the area.

**Answer D:** Pale color is a change associated with chronic somatic dysfunction. It is caused by vasoconstriction from prolonged hypersympathetic stimulation affecting blood vessels in the area.

**Answer E:** Prolonged blanching occurs with chronic somatic dysfunction because of vasoconstriction secondary to localized hypersympathetic activity. Prolonged blanching sometimes produces dermatographia or "skin writing", in which a patient will have prolonged erythema and sometimes urticaria where they were lightly touched or scratched. This patient has acute, not chronic, somatic dysfunction, so prolonged blanching will not be a finding.

## Acute vs Chronic Somatic Dysfunction

| Characteristic | Acute Somatic Dysfunction                                                 | Chronic Somatic Dysfunction                                         |
|----------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| History        | Recent injury                                                             | Injury $\geq$ 1 week before                                         |
| Pain           | Acute, severe, sharp, fire-like                                           | Dull, achy, itchy, "crawling", gnawing, burning                     |
| Temperature    | Increased (vasodilation)                                                  | Decreased (vasoconstriction)                                        |
| Skin           | Warm, moist, red/inflamed                                                 | Cool, dry, pale                                                     |
| Soft tissue    | Edematous, acute ("boggy") congestion                                     | Chronic congestion, stringy, ropiness, fibrotic, contracture        |
| Musculature    | Increased tone, leading to contraction, spasm, and poor quality of motion | Decreased tone, "mushy", limited range of motion due to contracture |

\*\*\*Pearl: Know this table as well! Try to understand the physiology to help you rather than just memorizing. For acute tissue texture changes – think of an acute injury like a sprained ankle vs chronic – think of the extreme which would be a patient with paralysis, cerebral palsy, post-stroke that will have those chronic tissue texture changes. They love to give you a mix of chronic and acute findings and then have you determine which might relate to a particular acute/new complaint vs problem in the PMH.

## TART

| Findings                                                | Acute                                                                                                                 | Chronic                                                                                                  |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Tissue texture changes<br>Muscle<br>Skin<br>Soft tissue | Increased tone, spasm<br><br>Warm, moist, red, inflamed<br><br>Boggy, edematous, fluid build up from vascular leakage | Decreased tone, flaccid, mushy<br><br>Cool, pale<br><br>Doughy, stringy, fibrotic, thickened, contracted |
| Asymmetry                                               | Present                                                                                                               | Present with compensation in other areas of the body                                                     |
| Restriction                                             | Present, painful with movement                                                                                        | Present, decreased or no pain                                                                            |
| Tenderness                                              | Greatest                                                                                                              | Present but less                                                                                         |
| Viscero-somatic reflexes                                | Uncommon or minimal                                                                                                   | Common                                                                                                   |

Important note - Classically, prolonged blanching with deep tissue texture assessment in lighter-skinned individuals is associated with chronic changes. In darker-skinned individuals, prolonged blanching can be associated with both acute and chronic changes.

Savarese OMT Review

\*\*\*This is a slide from Touro content for your review and additional supplementation.

# TART Changes

And another version!

| Characteristic     | Acute (hours to days)                                                                                                | Chronic (weeks to months)                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Temperature        | Warm                                                                                                                 | Cool                                                                                                                           |
| Moisture           | Moist                                                                                                                | Dry                                                                                                                            |
| Edema              | Present                                                                                                              | Absent                                                                                                                         |
| Tenderness         | Greatest                                                                                                             | Present, but less                                                                                                              |
| Skin Thickness     | Increased                                                                                                            | Decreased                                                                                                                      |
| Fibrosis           | Absent                                                                                                               | Present                                                                                                                        |
| Tissue Textures    | Boggy<br>Rough                                                                                                       | Stringy, Ropy<br>Smooth                                                                                                        |
| Skin Drag          | Sticky                                                                                                               | Slippery                                                                                                                       |
| Skin color changes | Hyperemic (red on lighter color skin due to increased blood flow, but varies based on skin tone)<br>Redness persists | Ischemic (blanched)<br>Redness (varies based on skin tone, red more common in lighter skin tones)<br>fades quickly or blanches |

Acute – sprained ankle

Chronic – CP vs CVA

Can have acute on chronic

## Question 4

A 17 y/o M presents with pain in his lower back after a Taekwondo tournament yesterday. He states that he was kicked in the lower back when the pain began and describes it as sharp muscle spasm that is worse with walking and bending. He denies radiating symptoms, burning pain, or LE weakness. When asked to point to the location of the pain, the pt points to the paraspinal regions bilaterally in the lower lumbar area. ROM testing reveals reduction in flexion and sidebending bilaterally at the level of the lumbar spine, but ROM in extension is preserved. A series of lumbar radiographs are performed and demonstrate no evidence of acute fractures. After a thorough PE, he is treated with an indirect OMT technique. Osteopathic palpatory exam of the lumbar region is most likely to reveal which of the following findings?



\*\*\*Pearl: Please note that this is very similar to question 3 – see a recurring theme here? 😊 Note that this is referring more to viscerosomatic vs acute MSK injury, but tissue texture changes the same!

## Explanation and Bottom Line

Bottom Line: acute viscerosomatic reflexes present with tissue texture changes described as increased muscle tone, boggy, edematous, increased temperature, vascular congestion, increased tenderness, moist skin, persistent erythema, and sharp pain.

The palpatory findings are different for an acute somatic dysfunction and a chronic **somatic dysfunction**. Tissue texture changes, asymmetry, restriction, and tenderness (TART) criteria are often used in determining somatic dysfunction. This patient has an acute injury that occurred 1 day ago and associated lumbar findings would be **acute** in nature, such as **bogginess**.

A viscerosomatic reflex occurs when localized visceral stimuli produce patterns of reflex response in segmentally related somatic structures. This continued visceral dysfunction transmits abnormal sensory input into the spinal cord, resulting in facilitation of the interneurons. This results in an exaggerated efferent response to somatic structures and the pain.

Tissue texture changes vary in their palpable quality based on chronicity. These quality differences are as follows.

- Acute findings: Increased muscle tone, boggy, edematous, increased temperature, vascular congestion, increased tenderness, moist skin, persistent erythema, and sharp pain
- Chronic findings: Ropy, stringy, doughy, flaccidity, diminished muscle tone, mushy feeling, fascial contracture, limited range of motion, cool temperature, dry skin, thickening, pimples present, and dull pain

# Distractors

**Answer B:** Cool, dry skin does not present with acute somatic dysfunction. Acute somatic dysfunctions would have warm, moist skin that could appear inflamed.

**Answer C:** Doughy, fibrotic tissues that are harder to palpate secondary to decreased blood flow because of vascular constriction are associated with chronic somatic dysfunctions. Acute somatic dysfunction, as seen in this patient, generally has boggy, edematous tissues secondary to vascular leakage and chemical reactions within the tissues, making this answer incorrect.

**Answer D:** Flaccid is consistent with chronic findings. This patient has an acute dysfunction and his muscles will demonstrate increased tone instead.

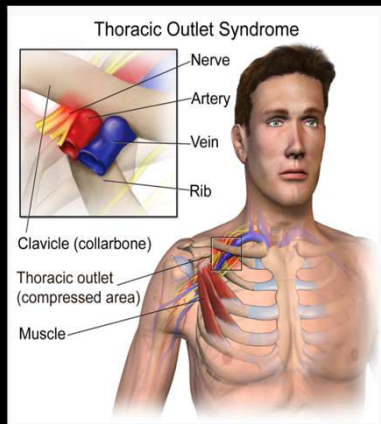
**Answer E:** Ropy is consistent with a chronic finding. This change was made in the 4th edition of *Foundations of Osteopathic Medicine* published in 2019. Ropy was previously classified as an acute finding in the 3rd edition. Ropy as a chronic finding is the correct classification for the COMLEX moving forward. **\*\*Important to note!\*\***

# Question 5

A 30 y/o F tennis player presents to the office with discomfort in her right shoulder, arm, and hand. She has no history of trauma. The patient complains of vague symptoms including pain, numbness, tingling, and cooling sensation in her entire right UE. Symptoms began when she started playing significantly more tennis on the weekends. A radiographic workup is negative for pathology in her shoulder, cervical spine, and lung apices. Structural exam reveals a restricted right sternoclavicular joint, an inhaled first rib, hypertonic anterior scalenes on the right, painful shoulder ROM, and some limited cervical spine AROM secondary to pain. Which of the following physical examination maneuvers would most likely be positive in this patient?

# Explanation and Bottom Line

Bottom line: Adson test involves the patient extending the elbow and extending, externally rotating, and abducting the shoulder before turning their head toward the ipsilateral arm. A positive test indicates the presence of thoracic outlet syndrome.



This patient is most likely suffering from **thoracic outlet syndrome**. Individuals may complain of vague symptoms that include pain, numbness, tingling, and cooling sensations of the affected upper extremity. This condition is most commonly caused by compression of the neurovasculature by the first rib or the scalene muscle group from excessive overhead activities.

Vascular thoracic outlet syndrome is rare and involves compression of the subclavian artery or vein. A purely neurogenic thoracic outlet syndrome involving compression of the brachial plexus is also rare. Other less common causes can occur from a cervical rib, somatic dysfunction in the upper ribs, or abnormal insertion of the pectoralis minor muscle. Radiographs of the shoulder and C-spine are typically normal, and a chest radiograph should be completed to rule out the possibility of an apical lung tumor.

On physical examination, these patients will have a positive **Adson test**, positive Wright test, and decreased pulses in the upper extremity. To perform Adson test, the patient's radial pulse is palpated, the arm is extended at the elbow, and the shoulder is extended, externally rotated, and abducted. Patients then turn their head toward the ipsilateral arm. A positive test occurs when the pulse significantly decreases or disappears.

## Distractors

**Answer B:** The Allen test evaluates blood supply to the hand. The patient makes a fist while the physician compresses both the radial and ulnar arteries at the wrist. The patient relaxes his or her hand, which should appear pale. Releasing pressure on one of the arteries results in flushing of the hand. This process is repeated with the other artery. If the hand flushes slowly or does not flush, then that artery is not adequately perfusing the hand.

**Answer C:** The Speed test evaluates the biceps tendon as it courses through the bicipital groove. To perform this test, the patient fully extends the elbow, flexes the shoulder, and supinates the forearm. The patient then flexes the shoulder while the physician provides resistance. A positive test occurs with pain in the bicipital groove, suggesting bicipital tendonitis.

**Answer D:** The Spurling test evaluates the presence of cervical foraminal narrowing. The physician extends the patient's C-spine and side-bends it toward the side being tested. Pressure is directed downward, compressing the cervical foramina. A positive test occurs when pain radiates into the arm.

**Answer E:** The Wallenberg test evaluates for vertebral artery insufficiency. The patient lies down in the supine position and the physician holds the patient's neck in flexion, extension, right rotation, left rotation, right sidebending, and left sidebending. A positive test occurs when the patient experiences dizziness, lightheadedness, blurred vision, or nystagmus in any of the positions listed.

| PELVIS & HIP           |                                                                                                                                                           |                                                        |                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| TEST                   | PROCEDURE                                                                                                                                                 | POSITIVE SIGN                                          | INTERPRETATION                                                                                  |
| Lumbosacral Spring     | Pt prone; with heel of hand over lumbosacral junction, attempt springing motion                                                                           | Minimal amount of spring                               | Sacral base is held posterior (e.g. RSE, L/NST, bilateral sacral extension)                     |
| Sphinx (backward bend) | Pt prone; palpate sacral sulcus or ILA bilat; Pt props up on elbows; note symmetry                                                                        | Landmarks become more asymmetrical                     | More asymmetrical landmarks= side of shallow sulcus sacral base is held posterior               |
| Ober's                 | Pt lateral recumbent, involved leg up; passively abduct leg with knee bent, slightly extend hip. Slowly release leg                                       | Thigh stays in abducted position                       | Contracture of tensor fascia lata or ITB                                                        |
| Thomas                 | Pt supine, hips & knees flexed; arms hug legs toward chest; actively drops one leg at a time, evaluate lumbar lordosis & ability to lie leg flat on table | Increased lordosis and/or space between knee and table | Psoas muscle contraction or contracture                                                         |
| Patrick (FABERE)       | Pt supine, Flex, Abduct, External Rotate, then extend hip (stabilize opposite ASIS)                                                                       | Pain, Decreased quality/quantity motion                | Hip dysfunction—pain; SI joint, hip joint dysfunction & hip muscle contraction—decreased motion |

| LOWER EXTREMITY                            |                                                                                                                                                                              |                                                     |                                                                                |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------|
| TEST                                       | PROCEDURE                                                                                                                                                                    | POSITIVE SIGN                                       | INTERPRETATION                                                                 |
| Apley Compression                          | Pt prone; Flex knee to 90°. Compress down thru heel; int & ext rotate tibia                                                                                                  | Knee pain                                           | Meniscus tear                                                                  |
| Apley Distraction                          | Pt prone; Flex knee to 90°. Distract pulling up on foot; int & ext rotate tibia                                                                                              | Knee pain                                           | Ligamentous injury (commonly MCL or LCL)                                       |
| Lachman's                                  | Pt supine; Grasp proximal tibia and distal femur; Flex knee 10°–30°, pull tibia forward                                                                                      | Tibia moves excessively out from under the femur    | ACL instability                                                                |
| McMurray's                                 | Pt supine; fully flex one leg, monitor medial knee joint line; rotate tibia to put valgus stress on knee; slowly extend knee. Repeat with varus stress (internal rotation)   | Palpable or audible click especially with knee pain | Posterior medial meniscus tear—valgus<br>Posterior lateral meniscus tear—varus |
| Anterior & Posterior Drawer (knee & ankle) | Pt supine; flex knee 90° with foot on table; completely stabilize foot; wrap hands around proximal tibia; pull anterior (anterior drawer); push posterior (posterior drawer) | Tibia slides or moves excessively under femur       | Anterior drawer=ACL tear<br>Posterior drawer=PCL tear                          |

| LOW BACK          |                                                                                                                                       |                                                                          |                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEST              | PROCEDURE                                                                                                                             | POSITIVE SIGN                                                            | INTERPRETATION                                                                                                                                                |
| Hip Drop          | Pt standing; monitor iliac crest heights; pt bend one knee, keep foot on floor                                                        | Ipsilateral hip elevates; unable to sidebend L—spine toward straight leg | Loss of lumbar sidebending                                                                                                                                    |
| Pelvic Side Shift | Pt standing; stabilize R shoulder while pushing pelvis toward R; repeat on L, side                                                    | Side of freer translation                                                | Psoas syndrome on R will translate to L but not to R                                                                                                          |
| Lasegue's (SLR)   | Pt supine; stabilize ASIS, raise leg (normal: raise to 80°); if leg raises to <80° (40° pain), lower slightly and dorsiflex the foot. | pain between 30°–80°                                                     | no pain with dorsiflexion = right hamstrings<br>pain with dorsiflexion = lumbar radiculopathy ± contralateral pain = herniated lumbar disc (highly sensitive) |
| Crossed Leg Raise | Asymptomatic leg is raised                                                                                                            | Increased pain in contralateral leg                                      | Contralateral test (CLR): highly specific (almost diagnostic of herniation)                                                                                   |
| Trendelenberg     | Pt standing; observe at level of iliac crests; Pt stands on one leg, by bending and lifting opposite foot off floor                   | Iliac crest height drops on non-stance side                              | Weak gluteus medius on weight-bearing side                                                                                                                    |
| Psoas Tension     | Pt prone; flex pt knee to 90°; hold pt thigh; extend hip; repeat on other side                                                        | Restriction of extension motion                                          | Iliopsoas tight on side of restricted motion                                                                                                                  |

| SHOULDER        |                                                                                                                                          |                                                             |                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------|
| TEST            | PROCEDURE                                                                                                                                | POSITIVE SIGN                                               | INTERPRETATION                                                   |
| Apley's Scratch | Pt touches superior & inferior aspects of opposite scapula                                                                               | Decreased range of shoulder motion                          | Rotator cuff dysfunction                                         |
| Empty Can       | Pt attempt to elevate arms against resistance; shoulder flexed to 90°, internal rotated, elbow extended, forearm pronated (thumbs down)  | Weakness                                                    | Rotator cuff tear; suprascapular nerve entrapment or neuropathy  |
| Drop Arm        | Passively abduct pt affected arm at least 90°; pt try to lower arm slowly                                                                | Arm drops without control                                   | Rotator cuff tear; supraspinatus weakness                        |
| Neer's          | Passively, fully flex pt arm & pronate forearm                                                                                           | Subacromial pain                                            | Subacromial impingement                                          |
| Hawkin's        | Passive flexion of shoulder 90°; elbow bent; internally rotate shoulder                                                                  | Subacromial pain                                            | Supraspinatus tendon impingement or rotator cuff tendinitis      |
| Speed's         | With mild flexion at elbow, flex pt shoulder 60° and supinate forearm; Pt attempts further flexion of shoulder; palpate bicipital groove | Pain; Lateral or medial movement of biceps tendon in groove | Biceps tendon instability or tendinitis                          |
| Yergason's      | Pt elbow flex to 90° forearm pronated; pt attempt supination and flexion of elbow against resistance                                     | Pain in biceps tendon (long head)                           | Biceps tendinitis or instability                                 |
| Apprehension    | Abduct pt arm to 90°; externally rotate & apply anterior pressure to humerus                                                             | Pain or apprehension about impending subluxation            | Anterior glenohumeral instability (chronic dislocating shoulder) |

| THORACIC OUTLET & CARPAL TUNNEL & WRIST |                                                                                                                                         |                                     |                                                                                                                          |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| TEST                                    | PROCEDURE                                                                                                                               | POSITIVE SIGN                       | INTERPRETATION                                                                                                           |
| Adson's                                 | Passively abduct & extend Pt arm (affected side); palpate pulse; pt turns head away & inhales deeply                                    | Diminished pulse                    | Compression/occlusion subclavian artery between anterior & middle scalene muscles ("Scalenes—antious Synd")              |
| Wright's                                | Hyperabduct pt arm above head; palpate radial pulse                                                                                     | Radial pulse severely diminished    | Compression of neurovascular bundle between pectoralis minor muscle and coracoclavicular process ("Hyperabduction Synd") |
| Costoclavicular (Military Posture)      | Passively depress and extend pt shoulder; monitor radial pulse                                                                          | Radial pulse severely diminished    | Compression of neurovascular bundle between clavicle & 1 <sup>st</sup> rib ("Costoclavicular Synd")                      |
| Phalen's                                | Pt flexes wrists; presses back of hands together; forearms horizontal; hold for one minute                                              | Pain & paresthesia of affected hand | Carpal tunnel syndrome                                                                                                   |
| Reverse Phalen's (prayer)               | Pt extends wrists; presses palms together; forearms horizontal; hold one minute                                                         | Pain & paresthesia of affected hand | Carpal tunnel syndrome                                                                                                   |
| Tinel's (Wrist)                         | Tap over the volar carpal ligament—distal to wrist crease                                                                               | Paresthesias in first 3 fingers     | Carpal tunnel syndrome                                                                                                   |
| Finkelstein                             | Pt make fist with thumb tucks inside; ulnar deviate wrist                                                                               | Pain in wrist/forearm               | deQuervain's disease (stenosing tenosynovitis of abductor pollicis longus & extensor pollicis brevis tendons)            |
| Allen                                   | Occlude both radial & ulnar artery at wrist; pt open & close fist until palm becomes pale; release one artery; repeat with other artery | Hand remains pale                   | Occlusion or poor function of released artery                                                                            |

| CERVICAL IMPINGEMENT |                                                                                     |                                                     |                                                                        |
|----------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------|
| TEST                 | PROCEDURE                                                                           | POSITIVE SIGN                                       | INTERPRETATION                                                         |
| Spurling             | Pt seated; extend neck; rotate head toward affected side; press down on top of head | Radicular pain or paresthesia in dermatomal pattern | Cervical nerve root impingement (for animal narrowing or inflammation) |

\*\*\*Found a great summary resource of commonly described orthopedic tests on COMLEX at:  
[https://www.reddit.com/r/comlex/comments/1di4r1q/common\\_msk\\_tests\\_on\\_comlex/#lightbox](https://www.reddit.com/r/comlex/comments/1di4r1q/common_msk_tests_on_comlex/#lightbox)

## PELVIS & HIP

### Spring Test (Lumbosacral Spring test)

Assesses whether or not sacral base is tilted posteriorly. Patient prone, physician places heel of hand over lumbosacral junction, attempting to spring at junction. Positive if there is little or no springing, indicating sacral base posterior. **Note:** Will be positive with a unilateral sacral extension, sacral margin posterior, sacral base posterior, and backward rotation on an oblique axis.



### Sphinx Test (Backward Bending Test)

Determines if sacral base has moved anteriorly or posteriorly. Patient prone, physician's thumbs in sacral sulci. If sulci depths differ, test will determine if one side has moved anteriorly, or if the other side has moved posteriorly. While monitoring sacral sulci, have patient prop up on their elbows. If depth difference increases, side of the shallow sulcus is being held posteriorly. If depth difference becomes more symmetric, side that was originally deeper is being held anteriorly.



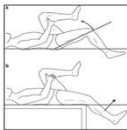
### Ober's Test

Tests for contracture of the tensor fascia lata or iliotibial band. Patient lies lateral recumbent on side opposite leg to be tested. Physician lifts leg into abduction with knee bent, slightly extending the hip. Slowly release leg. Thigh should drop towards table, not stay in abducted position.



### Thomas Test

Tests for contraction or contracture of the psoas muscle. Patient supine with hips and knees flexed, arms hugging legs towards chest. Instruct patient to drop one leg at a time while you evaluate for lumbar lordosis and space between table and knee of straightened leg. Positive test would reveal increased lordosis and/or space between knee and table on side of dysfunctional psoas.



### Patrick Test (aka FABERE Test)

Evaluates hip joint, sacroiliac joint, and surrounding tissues, especially osteoarthritis of the hip. Patient supine. Bring leg into Flexion (hip and knee), Abduction (knee is abducted, but foot stays near opposite knee), External Rotation, and Extension (push knee further towards the table while stabilizing contralateral ASIS). Notice quality and quantity of motion, and location of pain. Decreased motion may correspond to SI joint, hip joint, muscular contractions (e.g. adductors, iliopsoas). Any hip dysfunction will cause pain and inability to complete the test.



## LOWER EXTREMITY

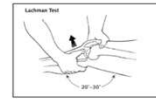
### Apley's Compression and Distraction Tests

Evaluates the menisci and ligamentous structures of the knee (most commonly medial and lateral collateral ligaments). Patient prone. Physician flexes knee to 90 degrees. While compressing straight down on the heel, physician internally and externally rotates the tibia. Pain indicates meniscal tear. Then physician pulls foot upward (distraction) and internally and externally rotates tibia. Pain indicates ligamentous injury.



### Lachman's Test

More accurate test for assessing stability of ACL. Patient supine. Physician grasps proximal tibia with one hand and distal femur with other hand, with knee flexed between 10-30 degrees. Pull tibia forward, noticing if tibia moves excessively out from under the femur.



### McMurray's Test

Evaluates for tears in the posterior aspect of the menisci. Patient supine. Physician fully flexes one leg, with one hand monitoring medial knee joint line. Physician rotates tibia externally and places a valgus stress on the knee. Maintain position while slowly extending the knee. A palpable or audible click, especially with pain, indicates posterior medial meniscus tear. Repeat test for posterior lateral meniscus tear by rotating tibia internally and placing a varus stress.



### Anterior and Posterior Drawer Tests

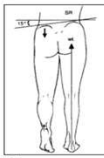
Tests for anterior and posterior cruciate ligament dysfunction, respectively. Patient supine. Flex knee to 90 degrees, foot on table. Physician sits on patient's foot to stabilize it. Physician wraps both hands behind tibia, pulls anteriorly (anterior draw, testing ACL). Positive if tibia slide out from under femur. The physician pushes posteriorly on tibia (posterior drawer, testing PCL). Positive if tibia excessively moves backward under the femur.



## LOW BACK

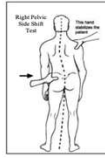
### Hip Drop Test

Assesses sidebending ability of lumbar spine and thoracolumbar junction. Patient standing, physician monitoring iliac crest heights. Patient bends one knee while keeping heel in floor. Normal response: lumbar spine sidebends towards contralateral side as bent knee, creating smooth convex curve in lumbar spine, and ipsilateral iliac crest drops >20-25 degrees.



### Pelvic Side Shift Test

Determines if sacrum is midline. Patient standing. Physician behind patient, stabilizes right shoulder while pushing pelvis towards right, then repeat on left side. Positive on side of freer translation. Note: Psoas syndrome on right will cause a positive test to the left.



### Lasague's Test (Straight Leg Raising Test)

Tests for possible lumbar radiculopathy.

**Seated:** Patient sits up straight, holding edge of table for support. Physician brings one leg into extension, then repeats on other side. Pain radiating down leg indicates possible radiculopathy. Inability of patient to stay seated straight up, or to fully extend knee, indicates contraction of hamstring muscles.

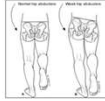
**Supine:** Patient supine. Physicians cephalad hand holds ASIS on side of leg being raised. Knee must remain straight. If ASIS moves superiorly indicating pelvic rotation, hamstrings are tight and will limit ability to test. If leg raising creates back/sciatic pain, drop leg slightly and dorsiflex ankle. If creates pain again, more indicative of a radiculopathy. If leg raising creates back/sciatic pain on the contralateral side, further evidence of space-occupying lesion (e.g. herniated disc) in lumbar area.



### Trendelenberg Test

Evaluates for gluteus medius weakness (also for dislocation of the hip, or a nonunion of the neck of the femur).

Patient stands. Physician kneels or squats behind patient, eyes at level of patient's pelvis. Instruct patient to stand on one leg, bending opposite hip and knee to lift foot from floor (physician may lightly hold patient's hips to help patient balance). Note any change in height of iliac crest on nonstance side. A weak gluteus medius on the weight-bearing side will cause the opposite hip to drop (normally, that side will rise).



### Psoas Tension

The tension of the psoas muscles, which play a role in the creation and maintenance of lumbar segmental somatic dysfunctions, can be assessed as follows: Patient prone on examination table. Practitioner stands at either side of table. Practitioner flexes the patient's knee to 90 degrees and grasps the patient's thigh just above the knee with one hand. Practitioner extends the patient's hip until the ASIS on that side rises off the table. Repeat with opposite leg. Practitioner observes the range and quality of the hip extension motion on each side. Restriction in the range or quality of motion on either side indicates tightness of the iliopsoas muscle on that side.



## SHOULDER

### Apley Scratch Test

Evaluates range of shoulder motion. Patients reach behind their head to touch superior medial angle of opposite scapula (abduction/external rotation). Can also have patient reach across chest to touch opposite shoulder (adduction/internal rotation). Then have patient reach behind back up towards inferior angle of opposite scapula (more intense adduction/internal rotation).



### Empty Can Test

Evaluates for Supraspinatus tendonitis. Shoulders abducted to 90 degrees, 30 degrees towards the front of the body, arm internally rotated (thumbs pointing downward). Physician gently pushes down on arms. Pain at shoulder indicates a positive test.



### Drop arm Test

Tests for rotator cuff tear. Abduct patient's arm to 90 degrees, then release arm. If the arm drops to the side, there is a tear. If a partial tear is present, the arm may remain up, but will drop when you tap on the arm.



### Neer's Test

Tests for impingement of rotator cuff tendons under the coracoacromial arch. Patient's arm is placed in forced flexion with the arm fully pronated. The scapula should be stabilized during the maneuver to prevent scapulothoracic motion. Pain with this maneuver is a sign of subacromial impingement.



### Hawkins' Test

Tests for subacromial impingement or rotator cuff tendonitis. Physician elevates patient's arm forward to 90 degrees while forcibly internally rotating the shoulder, elbow slightly bent. Pain suggests subacromial impingement or rotator cuff tendonitis.



### Speed's Test

Tests for pathology of long head of bicep in the bicipital groove. With patient's elbow extended and forearm supinated, physician resists shoulder flexion. Positive test with tenderness in bicipital groove.



### Yergason's Test

Patients with rotator cuff tendonitis frequently have concomitant inflammation of the biceps tendon. The Yergason test is used to evaluate the biceps tendon. In this test, the patient's elbow is flexed to 90 degrees with the thumb up. The examiner grasps the wrist, resisting attempts by the patient to actively supinate the arm and flex the elbow. Pain with this maneuver indicates biceps tendonitis.



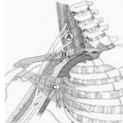
### Apprehension Test

Tests for a chronically dislocating shoulder. Abduct, extend, and externally rotate shoulder. Patient will appear apprehensive at the point where the shoulder is about to dislocate. (If someone is aware that their shoulder dislocates, they will usually mention this in the history)



### THORACIC OUTLET SYNDROMES

#### A: Adson's Test



Evaluates for compromise of the neurovascular bundle due to a tight anterior scalene muscle, a cervical rib, or a pancoast tumor. Monitor patient's radial pulse while extending elbow and shoulder, externally rotating and slightly abducted. Instruct patient to inhale deeply and turn their head towards ipsilateral arm. Positive if radial pulse severely diminishes. If negative, this test can be repeated with the head turned to the opposite side.



#### B: [Eden's Test] Costoclavicular Syndrome/Military Posture Test

Evaluates for compromise of neurovascular bundle between clavicle and first rib. Monitor radial pulse while depressing and extending the shoulder. Positive if radial pulse severely diminishes.



#### C: Wright's Test

Evaluates for compromise of neurovascular bundle passing under pectoralis minor muscle at coracoid process. Monitor radial pulse while hyperabducting arm above head. Positive if radial pulse severely diminishes.



### CARPAL TUNEL SYNDROME & WRIST

#### Phalen's Test

Tests for carpal tunnel syndrome. Patient flexes wrists and presses backs of hands together. Forearms must be horizontal. Hold for one minute. A positive test will produce pain and/or paresthesias in the affected hand.



#### Prayer Test (Reverse Phalen)

As for Phalen's, except palms of hands are together, and wrists are fully extended. Forearms must be horizontal. Alternatively, patient can push hand (wrist extended) against physician's hand while gripping physician's hand.



#### Tinel's Sign (Wrist and Elbow)

**Wrist:** Tap over the volar carpal ligament—just distal to distal wrist crease. Paresthesias in the first three (and lateral half of 4<sup>th</sup>) fingers indicate carpal tunnel syndrome.



**Elbow:** Tap the area of the ulnar nerve between the olecranon and the medial epicondyle. Positive test will cause paresthesias down the forearm and to the ulnar distribution of the hand. Indicates ulnar nerve entrapment at the elbow. **Note:** May also use Tinel's test at fibular head for peroneal compression, and at the ankle for posterior tibial nerve entrapment.



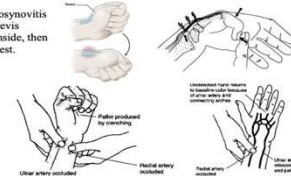
#### Finkelstein Test

Evaluates for de Quervain's disease (stenosing tenosynovitis of abductor pollicis longus and extensor pollicis brevis tendons). Patient makes a fist with thumb tucked inside, then ulnar deviates the wrist. Pain indicates a positive test.



#### Allen Test

Evaluates functioning of the medial and ulnar arteries. Occlude each artery at the wrist and have patient open and close their fist. Palm should become pale. Hand should flush with release of one artery. Repeat with the other artery.



#### Spurling Test (Compression Test)

Tests for neural foramina narrowing. Patient seated. Physician extends and sidebends head/neck to side being tested, and pushes down on head. Positive test is pain radiating into ipsilateral arm (with affected nerve root revealed in pain distribution). Pressure on facet joints or muscle spasm may also cause pain.



### CERVICAL IMPINGEMENT

## Question 6

54 y/o M presents to the office with complaint of difficult urination. History reveals he has problems initiating and maintaining a steady stream. Physical examination reveals no suprapubic tenderness. Digital rectal exam reveals a large, smooth, boggy prostate gland. Which of the following additional findings are most likely?

## Explanation and Bottom Line

- Bottom line: The Chapman's point for the prostate is located anteriorly along the posterior margin of the iliotibial band.

This patient is most likely suffering from **benign prostatic hyperplasia (BPH)**, characterized by difficulty initiating and maintaining a steady urine stream, increased urge or frequency of urination, and nocturia. Carcinoma of the prostate should certainly be considered, but the smooth nature of the prostate on examination makes that diagnosis less likely.

The **anterior Chapman point for the prostate is located along the posterior margin of the iliotibial band** and should be present with prostate pathology. Remember that the iliotibial band is a thick band of fibrous tissue that runs along the lateral aspect of the thigh. Note that the Chapman point for the prostate is different from the sympathetic innervation to the prostate, which occurs from T12/L1 to L2, and would be the expected location for a viscerosomatic reflex.

# Distractors

---

**Answer A:** L4 is not associated with any autonomic innervation and would therefore be unrelated to prostate pathology.

**Answer B:** Dysfunction of T9 could easily be associated with dysfunction of the upper gastrointestinal (GI) tract due to sympathetic connection, but it is not specifically associated with prostate pathology. T5-9 generally provides sympathetic innervation of the gastrointestinal tract to the ligament of Treitz. Organs included in this region are the distal esophagus, stomach, liver, gallbladder, proximal duodenum, spleen, and portions of the pancreas.

**Answer C:** Warm, taut paraspinal musculature at T3-T4 would better coincide with heart or lung pathology as evidence of elevated sympathetic tone. The sympathetic viscerosomatic reflex for thoracic viscera arises in segments T1-T5.

**Answer E:** The Chapman point for the urethra and/or the ovaries is located at the superior margin of the pubic ramus. Its presence would be less likely than the prostate point. Signs of urethral disease, either a sexually transmitted infection or other urinary infection, would more likely present with urethral discharge, fever, suprapubic pain, and increased urinary frequency, which are not seen in this case.

## Insight

Chapman points are a must-know for the COMLEX. These are easy points and should not be overlooked while studying. This question follows a very common COMLEX format, combining the concepts of sympathetic innervation and Chapman points, and forcing you to distinguish their relationship to an organ. Look for one like this on test day.

## GU Chapman's Points

The table below summarizes the GU Chapman points.

| Genitourinary Chapman Points |                                                                     |                                       |
|------------------------------|---------------------------------------------------------------------|---------------------------------------|
| Organ                        | Anterior                                                            | Posterior                             |
| <b>Adrenals</b>              | 1 inch lateral and 2 inches superior to the umbilicus (ipsilateral) | T11–T12 (ipsilateral)                 |
| <b>Kidneys</b>               | 1 inch lateral and 1 inch superior to the umbilicus (ipsilateral)   | T12–L1 (ipsilateral)                  |
| <b>Bladder</b>               | Periumbilical area                                                  | L2 TP bilaterally                     |
| <b>Ovaries/testes</b>        | Superior pubic ramus, 2 cm lateral to the symphysis                 | T9–T11 (ipsilateral)                  |
| <b>Urethra</b>               | Pubic symphysis                                                     | L2 TP bilaterally                     |
| <b>Prostate</b>              | Outer femur (posterior iliotibial band)                             | Lateral sacral base bilaterally       |
| <b>Uterus</b>                | Inferior pubic rami                                                 | L5 TP bilaterally, medial to the PSIS |
| TP = transverse process.     |                                                                     |                                       |

\*\*\*Another chart to know! Common ones I hear used more often are the adrenals vs kidneys vs gonads since they all involve measurements. Be able to differentiate those. Prostate is another that commonly shows up given some proximity to the GI colon points so they will try to confuse you with that distractor.

## Question 7

47 y/o M presents to the office with upper back pain for the past 1 week. The pain began when he fell asleep while sitting upright in a chair. On exam, it is noted that T7 is flexed, rotated to the right, and sidebent to the right. When palpating the transverse process of T7, you expect it to be located:

## Explanation and Bottom Line

- Bottom Line: Inferior angle of the scapula is located at the level of the spinous process of T7 and the TP of T8. This correlates with the rule of 3s, which states that the TP of T7-T9 are located 1 full segment above their corresponding SP. Therefore, the TP of T7 would be palpated 1 segment above the inferior angle of the scapula.

The spinous process of T7 is located at the level of the inferior angle of the scapula. This is also the location of the transverse process of T8, as the thoracic rule of 3s states that the spinous processes of T7, T8, and T9 are located 1 full segment below their transverse processes (in other words, the transverse processes are located 1 full segment above the corresponding spinous processes). Therefore, to palpate the **transverse processes of T7**, it would be necessary to move **1 segment above the level of the spinous process of T7, which is located at the level of the inferior angle of the scapula**.

To accurately diagnose, document, and treat somatic dysfunction, it is important to know basic anatomic landmarks. The following are useful landmarks for various structures:

- Vertebra prominens: C7 vertebra
- Spine of the scapula: T3 level
- Inferior angle of the scapula: level of the spinous process of T7 and the transverse process of T8
- Sternal notch: T2 level
- Sternal angle (angle of Louis): level of the 2nd pair of ribs and the spinous process of T4
- Iliac crest: level of L4 in males and L5 in females (because of their wider pelvis)
- Dimple of Venus: site of the posterior superior iliac spine; S2 level

Accurate diagnosis also relies on knowing the thoracic rule of 3s, which describes the relationships between the spinous processes and transverse processes in the thoracic spine. Thoracic vertebrae have a downslowing spinous process; therefore, the spinous process is lower than the corresponding transverse process for many of these vertebrae.

# Rule of 3s

| Thoracic Rule of 3s                            |                               |
|------------------------------------------------|-------------------------------|
| Segment                                        | Rule                          |
| T1–3                                           | SP is at the same level as TP |
| T4–6                                           | SP is 1/2 segment below TP    |
| T7–9                                           | SP is 1 segment below TP      |
| T10                                            | SP is 1 segment below TP      |
| T11                                            | SP is 1/2 segment below TP    |
| T12                                            | SP is at the same level as TP |
| SP = spinous process; TP = transverse process. |                               |

# “Rule of Threes”

Spinous processes in T-spine are long and variable

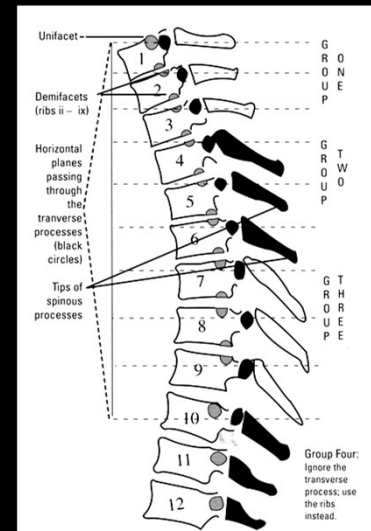
T1-T3 TPs at same level as SP

T4-T6 TPs halfway between SP of segment and one above

T7-T9 TPs one segment above SP

T10-T12 regresses:

- T10 one segment above
- T11 halfway between
- T12 level with TPs



\*\*\*Again know these in relation to T3 at spine of scapula, T7 at inferior angle, etc as this is a great way to test landmarks and rule of 3s.

# Distractors

---

**Answer B:** The vertebra prominens is another name for the C7 vertebra. This vertebra has an unusually long spinous process, which makes the vertebra especially prominent. It can be differentiated from T1, which also has a prominent spinous process, by palpating the spinous process during passive flexion and extension of the cervical spine. If the spinous process is part of C7, it will move during these movements; if it remains fixed, the spinous process is part of T1. The vertebra prominens is several segments above T7, so this answer is incorrect.

**Answer C:** The inferior angle of the scapula is located at the level of the spinous process of T7 and the transverse process of T8.

**Answer D:** The spine of the scapula is adjacent to the T3 vertebra. When diagnosing or treating abnormalities at T3, it is easily located by palpating the spine of the scapula and moving medially to the T3 transverse process.

**Answer E:** The sternal angle is the angle formed by the manubrium and the sternal body. It is located at the level of T4 spinous process. The sternal angle is the area where the costal cartilages of the 2nd pair of ribs articulate with the sternum.

## Question 8

56 y/o M presents to the office with complaints of blood visible in the toilet bowl with recent bowel movements. In addition, he reports blood that is visible on the toilet paper, along with an anal burning sensation when he wipes. Medical history is significant for internal hemorrhoids treated with rubber band ligation 5 years ago. A colonoscopy at age 52 was negative for polyps or diverticulosis. He admits to intense anal pruritic at night time and has found some improvement with an over the counter topical cream. During osteopathic exam, there is a 3 mm, fluid-filled rubbery nodule that is noted on the inner thigh region over the lesser trochanter. Where would you expect to find an associated Chapman point used to treat this diagnosis?

\*\*\*note that even if you haven't learned specific disease process, you know the anatomy and you know your ABCs so stick to those to help!

## Explanation and Bottom Line

- Bottom Line: Chapman points represent the somatic manifestation of a visceral dysfunction and are described as rubbery, discrete, palpable nodules approximately 2-3mm in diameter. The rectum's anterior point is found on the proximal inner thigh at the lesser trochanter. The rectum's posterior point is found on the sacrum at S2.

This patient's symptoms of blood in the toilet bowl during bowel movements, blood on the toilet paper when wiping, anal pruritus, and an anterior Chapman point on the lesser trochanter are consistent with a diagnosis of external hemorrhoids. With an anterior Chapman point for the rectum at the lesser trochanter, the associated **posterior Chapman point used for treatment would be found on the sacrum at the level of S2.**

Chapman points represent the somatic manifestations of a visceral dysfunction and are described as rubbery, discrete, palpable nodules approximately 2-3 mm in diameter. Their specific location corresponds with specific viscera, and therefore, they can assist in diagnosis.

Important differences between anterior and posterior points are discussed below.

- Anterior Chapman points are somatic manifestations and used to aid in making a diagnosis. They are located within the deep fascia or on the periosteum of the bone.
- Posterior Chapman points are more often used for treatment and located between the spinous and transverse processes of the vertebrae.

Palpation of a Chapman reflex will cause tenderness which is described as sharp in nature and nonradiating.

# GI Chapman Points

| Gastrointestinal Chapman Points   |                                               |                                                           |
|-----------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| Organ/Structure                   | Anterior                                      | Posterior*                                                |
| Esophagus                         | 2nd ICS                                       | T2–T3 bilaterally                                         |
| Stomach (acidity)                 | 5th ICS on the left                           | T5 on the left                                            |
| Stomach (peristalsis)             | 6th ICS on the left                           | T6 on the left                                            |
| Liver                             | 5th–6th ICS on the right                      | T5–T6 on the right                                        |
| Gallbladder                       | 6th ICS on the right                          | T6 on the right                                           |
| Pancreas                          | 7th ICS on the right                          | T7 on the right                                           |
| Spleen                            | 7th ICS on the left                           | T7 on the left                                            |
| Small bowel                       | 8th, 9th, 10th ICS bilaterally                | T8–T10 bilaterally                                        |
| Appendix                          | Tip of 12th rib on the right                  | T11–T12 on the right                                      |
| Cecum                             | Upper 1/5 of the right anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Ascending colon                   | Middle 3/5 of the right anterior thigh        | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Right 1/2 of the transverse colon | Lower 1/5 of the right anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Left 1/2 of the transverse colon  | Lower 1/5 of the left anterior thigh          | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Descending colon                  | Middle 3/5 of the left anterior thigh         | Triangular area between the L2 TP, L4 TP, and iliac crest |
| Rectum                            | Proximal inner thigh on the lesser trochanter | Sacrum at the level of S2                                 |

ICS = intercostal space; TP = transverse process.  
 \* All points from the esophagus to the appendix are located midway between the tip of the TP and the spinous process.

## Distractors

**Answer A:** The left transverse process of T7 is the posterior Chapman point to treat the spleen. This patient had no findings of splenic visceral dysfunction, making this choice incorrect.

**Answer C:** The T11-T12 intertransverse space is the posterior Chapman point used to treat the adrenals. This does not correspond to the rectum, where this patient's symptoms are presenting. Therefore, this is not the best answer.

**Answer D:** The transverse process of L2 is the posterior Chapman point used to treat the bladder. However, this patient is presenting with findings of anal pruritus and bright red blood per rectum, both of which are consistent with the diagnosis of external hemorrhoids. Therefore, a Chapman point on the sacrum at the level of S2 would be the best answer.

**Answer E:** The transverse process of T10 is the posterior Chapman point used to treat the ovaries/testes. This patient's diagnosis of external hemorrhoids is found in the rectum, corresponding with a posterior Chapman point found on the sacrum at the level of S2.

## Question 9

58 y/o M presents to the office with pain and paresthesias in his left anterolateral thigh. He is employed as a truck driver and sits for 8-12 hours at a time. The symptoms primarily occur when he is at work and improve with standing. PE reveals lumbar paravertebral hypertonicity and flattening of the lumbar lordosis. There is a tender point lateral to the AILS on the left. You decide to use counterstrain treatment for his somatic dysfunction. With the patient placed supine and his hips flexed, the most appropriate positioning is:

## Explanation and Bottom Line

- Bottom Line: Classic positioning for counterstrain treatment of the AL tenderpoints AL2, AL3, and AL4 involved the physician standing on the opposite side of the table and the pt placed supine with the knees and hips flexed to 90 degrees, ankles sidebent away from the tender point, and torso rotated toward the tender point.

This patient has a **lower extremity somatic dysfunction** with a tender point lateral to the anterior inferior iliac spine (AIIS), which is the location of the anterior lumbar 3 (AL3) tender point. The classic positioning for treating AL2, AL3, and AL4 tender points on the left involves placing the patient supine with the **hips and knees flexed to 90 degrees, ankles sidebent to the right** (away from the tender point), and **torso rotated to the left** (toward the tender point). The **physician stands on the right** (side opposite the tender point).

### Counterstrain Position of AL2 - AL4



Tender points occur after an injury or event that produces rapid shortening in a region, usually accompanied by lengthening of the opposing tissues. The body tries to prevent damage by contracting the myofascial tissues, resulting in hypershortening of the agonist muscles and tissues. This produces a lengthening response in the antagonists, ultimately resulting in hypertonic tissue, restricted motion, and a tender point.

\*\*\*Commonly see this group of points tested together since such similar positioning between them so know which point is medial, lateral, and inferior to the AIIS.

# AL Counterstrain Points

| Anterior Lumbar Tender Points and Treatment |                                                                       |                                                                                                                                                                                 |
|---------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tender Point                                | Location                                                              | Counterstrain Positioning                                                                                                                                                       |
| AL1                                         | Medial to the anterior superior iliac spine                           | <ul style="list-style-type: none"> <li>Physician: standing, same side as the tender point</li> <li>Patient: hips flexed, ankles sidebent toward, torso rotated away</li> </ul>  |
| AL2                                         | Medial to the AIIS                                                    | <ul style="list-style-type: none"> <li>Physician: standing, side opposite the tender point</li> <li>Patient: hips flexed, ankles sidebent away, torso rotated toward</li> </ul> |
| AL3                                         | Lateral to the AIIS                                                   | <ul style="list-style-type: none"> <li>Physician: standing, side opposite the tender point</li> <li>Patient: hips flexed, ankles sidebent away, torso rotated toward</li> </ul> |
| AL4                                         | Inferior to the AIIS                                                  | <ul style="list-style-type: none"> <li>Physician: standing, side opposite the tender point</li> <li>Patient: hips flexed, ankles sidebent away, torso rotated toward</li> </ul> |
| AL5                                         | Anterior superior aspect of the pubic ramus, lateral to the symphysis | <ul style="list-style-type: none"> <li>Physician: standing, same side as the tender point</li> <li>Patient: hips flexed, ankles sidebent away, torso rotated away</li> </ul>    |

## Distractors

**Answer A:** Physician standing on the left with the patient supine, hips flexed, ankles sidebent to the left, and torso rotated to the right describes the proper counterstrain treatment positioning for a left AL1 tender point. The AL1 tender point is located medial to the anterior superior iliac spine. This patient's tender point is lateral to the AILS, indicative of an AL3 tender point.

**Answer B:** Physician standing on the left with the patient supine, hips flexed, ankles sidebent to the right, and torso rotated to the right describes the classic counterstrain treatment positioning for a left AL5 tender point. The AL5 tender point is located at the anterior superior aspect of pubic ramus, lateral to the symphysis. In contrast, this patient's tender point is lateral to the AILS, indicative of an AL3 tender point.

**Answer C:** Physician standing on the right with the patient supine, hips flexed, ankles sidebent to the left, and torso rotated to the left describes the appropriate counterstrain treatment positioning for a right AL5 tender point. The AL5 tender point is located at the anterior superior aspect of pubic ramus, lateral to the symphysis. This patient's tender point is AL3, as it is lateral to the AILS.

**Answer D:** Physician standing on the right with the patient supine, hips flexed, ankles sidebent to the left, and torso rotated to the right is not the correct positioning for counterstrain treatment of any AL tender point.

## Question 10

62 y/o M presents to the office with a 3-week history of mild-to-moderate periorbital/frontal headache that is increasing in frequency. He has a new project at work and is on the computer for longer hours than usual. PE reveals protracted shoulders, head forward carriage, and a tender point on the inferior nuchal line immediately lateral to the midline. The most appropriate starting position for treating the tender point of the C spine with counterstrain is:

\*\*\*Did you catch the findings that should make you start thinking about Upper Crossed Syndrome?\*\*\*

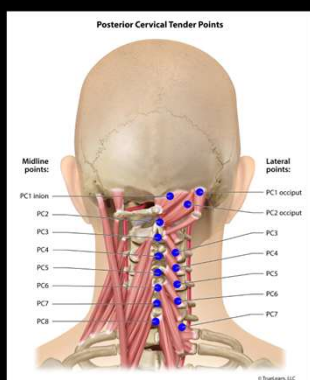
## Explanation and Bottom Line

Bottom Line: most posterior C spine midline tender points are treated in counterstrain with an extension, sidebend away, rotation away position. The PC1 inion, however, is a maverick tender point, and counterstrain treatment is primarily flexion of the OA joint.

This patient has a cervicogenic headache, which can result from upper crossed syndrome, as indicated by this patient's poor posture, protracted shoulder, forward head carriage, and long hours of sitting at a computer. This syndrome also causes somatic dysfunction of the cervical spine and results in the development of tender points, most specifically the **PC1 inion**. This point is located on the inferior nuchal line, immediately lateral to the inion. Counterstrain treatment of the PC1 inion is classically **marked flexion of the occipitoatlantal (OA) joint** with further refining of sidebending and rotation depending on the patient.

Most posterior cervical spine midline points are treated in an extension, sidebend away, rotate away (E Sa Ra) position. However, the PC1 inion point is one of the few points treated differently (a maverick tender point).

# PC Counterstrain Points



| Posterior Cervical Tender Points |                                                                                   |                                                                                  |
|----------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Tender Point                     | Location                                                                          | Treatment Position                                                               |
| PC1 inion                        | Inferior nuchal line, just lateral to the inion                                   | Flexion of OA (possibly minor sidebending toward, rotation away)                 |
| PC1 occiput                      | Inferior nuchal line midway between the inion and mastoid on the splenius capitis | Extension of OA, mild compression of head, slight sidebending, and rotation away |
| PC2 midline spinous process      | Superior or superior lateral tip of the C2 spinous process                        | Extension of OA, mild compression of head, slight sidebending, and rotation away |
| PC2 occiput                      | Inferior nuchal line within the semispinalis capitis muscle                       | Extension of OA, mild compression of head, slight sidebending, and rotation away |
| PC3 midline spinous process      | Inferior or inferolateral tip of the C2 spinous process                           | Flexion, sidebend away, rotation away                                            |
| PC3-PC7 lateral                  | Posterolateral tip of the articular process on the dysfunctional segment          | Extension, slight sidebending, and rotation away                                 |
| PC4-PC8 midline spinous process  | Inferior or inferolateral tip of the spinous process above the segment            | Extension, slight sidebending, and rotation away                                 |

\*\*\*Pearl: they love to test on the exceptions to the rules for counterstrain – PC1 inion, AC7 (different position). Also to help you learn counterstrain – pay attention to patterns – anterior tend to involve treating with flexion, posterior trend to involve treating with extension. In a pinch, try to find the point on yourself and see how you position yourself to treat.

## Distractors

**Answer A:** Extension by itself is not a typical position for any posterior cervical tender point, making it a poor choice in this patient. Most posterior cervical points are lateralized and then treated with a combination of extension plus sidebending and rotation.

**Answer C:** Rotation away is the counterstrain treatment for an AC1 tender point located on either the mandible or the C1 transverse process. However, this patient has a tender point on the inferior nuchal line immediately lateral to the midline, which is consistent with a PC1 point and requires neck flexion.

**Answer D:** Sidebend away and rotation away is the setup for counterstrain treatment of a PC1 tender point on the occiput. Some sources indicate the need for mild compression on the head to reduce the myofascial tension on tissues in the suboccipital region. The PC1 occipital tender point is located on the inferior nuchal line midway between theinion and mastoid. This patient's tender point is closer to the midline at theinion.

**Answer E:** Sidebend toward and rotation away can be the correct minor motion adjustments used for the counterstrain treatment of a posteriorinion point. However, flexion is the main movement for treating posteriorinion, making it the most appropriate choice for this patient.

# Question 11

29 y/o F presents to the ED with severe right flank pain and hematuria. KUB radiograph reveals a radiopaque, spherical object in the proximal ureter. Which osteopathic technique will be most effective for treating the parasympathetic viscerosomatic reflex in this patient?

## Explanation and Bottom Line

Bottom Line: Parasympathetic innervation of the kidneys and proximal ureters occurs via the vagus nerve, which originates in the medulla and passes close to the OA joint, C1, and C2. Suboccipital inhibition is effective for treating parasympathetic viscerosomatic reflexes from organs innervated by the vagus.



This patient likely has acute nephrolithiasis, as indicated by her flank pain, hematuria, and stone (radiopaque, spherical object) in the proximal ureter. **Suboccipital inhibition** is a myofascial release technique used to treat parasympathetic viscerosomatic reflexes in patients with kidney or proximal ureter pathology.

The kidneys and proximal ureters receive parasympathetic innervation from the vagus nerve. The vagus nerve originates in the medulla, exits the skull at the jugular foramen, and passes close to the occipital-atlanto joint, C1 vertebra, and C2 vertebra. It then travels in the carotid sheath inferiorly in the neck toward the thorax. The ureters receive dual parasympathetic innervation, with the proximal portion innervated by the vagus nerve and the distal portion innervated by the pelvic splanchnic nerves (S2-S4). Sympathetic innervation of the proximal ureters and kidneys originates from the T10-L1 vertebral levels.

Suboccipital inhibition is effective for treating parasympathetic viscerosomatic reflexes from organs innervated by the vagus. As shown in the image below, the technique is performed with the patient in the supine position. The physician's finger pads are placed just inferior to the superior nuchal line. The physician then uses the fingers to apply gentle, constant pressure to elevate the patient's head, while also applying cephalic traction. The head will slowly descend toward the table as the suboccipital muscles relax.

\*\*\*Pearl: note that you may see this technique called suboccipital release or inhibition. Know which techniques are best to address autonomics – sympathetics (rib raising, paraspinal inhibition) vs parasympathetics (suboccipital for vagus) vs sacral inhibition (S2-S4).

## Distractors

**Answer A:** Sacral rocking is a myofascial release technique for normalizing parasympathetic tone in patients with somatic dysfunction of the distal transverse colon, descending colon, sigmoid colon, rectum, distal ureter, bladder, and reproductive organs. As this patient has a proximal ureter stone, her parasympathetic viscerosomatic reflex arises from the vagus nerve, for which sacral rocking would not be effective.

**Answer C:** The T1-T4 (or T1-T5) rib raising technique is used to treat sympathetic viscerosomatic reflexes from the head, neck, heart (T1-5), and lungs (T2-6). As the technique treats sympathetic reflexes, it will not be effective for any parasympathetic viscerosomatic reflex.

**Answer D:** The T5-T9 rib raising technique treats sympathetic viscerosomatic reflexes from the foregut (distal esophagus, stomach, duodenum, spleen, gallbladder, and liver). This technique is not useful for any parasympathetic viscerosomatic reflex.

**Answer E:** The T10-T12 rib raising technique is used for treating sympathetic viscerosomatic reflexes from multiple pelvic and abdominal organs. It does not affect any parasympathetic reflex.

## Question 12

25 y/o M is in the hospital after undergoing an open appendectomy 2 days ago. He has had no bowel movements since before surgery. PE reveals a mildly distended abdomen and no bowel sounds, with a well-appearing abdominal incision site. Given the likely diagnosis, the most appropriate treatment for this patient is:

\*\*\*Pearl: you will have a hospital/surgical patient lab in the spring, but even without that, you can use your ABCs and some common sense (caution touching an abdomen directly that just had surgery on it!) to get an answer.

## Explanation and Bottom Line

Bottom Line: Paraspinal inhibition of the TL junction is a good treatment option for postoperative ileus. It not only reduces sympathetic hyperactivity of the bowel to improve peristalsis, but it also requires minimal patient positioning, making it a generally well-tolerated treatment in postoperative patients.

This patient is likely experiencing **postoperative ileus** given his abdominal distension, absent bowel sounds, lack of bowel movements, and recent surgery. Postoperative ileus is characterized by reduced or absent peristalsis for a prolonged period of time after surgery. **Paraspinal inhibition of the thoracolumbar junction** is the most appropriate osteopathic treatment for the patient at the current time.

Postoperative ileus involves mechanical, chemical, and autonomic nervous system-mediated processes. Manipulation of the bowel releases substance P and nitric oxide, leading to upregulation of the sympathetic nervous system and/or downregulation of the parasympathetic nervous system. Both nervous system changes can contribute to postoperative ileus. Spinal/skeletal rigidity occurs because of viscerosomatic reflexes, incisional pain, and surgical stress, resulting in maintained hypersympathetic/hypoparasympathetic activity.

A variety of treatment options are available to alleviate this patient's symptoms and facilitate peristalsis. These include treating the mesentery with myofascial release to resolve any restrictions involving the intestines, mesenteric ganglion release or thoracolumbar treatments to reduce sympathetic tone, or sacral treatment to increase parasympathetic tone. When selecting a treatment, always consider which treatment the patient would be able to tolerate and modify the technique as needed. Paraspinal inhibition of the thoracolumbar junction would require the least amount of patient positioning and manipulation at/near the surgical site, making it the treatment most likely to be well-tolerated by a postoperative patient.

The colon receives its sympathetic innervation from levels T10 to L2 (middle and lower gastrointestinal tract, as shown in the table below). Applying pressure to the paraspinal tissues at T10–L2 will help reduce hypersympathetic tone, thereby restoring normal sympathetic-parasympathetic balance. This will promote peristalsis and enable the patient to pass stool and gas.

# Viscerosomatic Reflexes

| Sympathetic Viscerosomatic Reflexes |                   |                 |                   |
|-------------------------------------|-------------------|-----------------|-------------------|
| Organ/Region                        | Spinal Cord Level | Organ/Region    | Spinal Cord Level |
| Head and neck                       | T1–T4             | Kidneys         | T10–T11           |
| Heart                               | T1–T5 (left)      | Upper ureters   | T10–T11           |
| Respiratory system                  | T2–T7             | Lower ureters   | T12–L1            |
| Esophagus                           | T2–T8             | Bladder         | T11–L2            |
| Upper GI tract                      | T5–T9             | Gonads          | T10–T11           |
| Middle GI tract                     | T10–T11           | Uterus/cervix   | T10–L2            |
| Lower GI tract                      | T12–L2            | Erectile tissue | T11–L2            |
| Appendix/cecum                      | T10–T12           | Prostate        | T12–L2            |
| Arms                                | T2–T8             | Legs            | T11–L2            |

Upper GI tract: stomach (left), liver and gallbladder (right), spleen (left), portions of the pancreas and duodenum  
Middle GI tract: portions of the pancreas and duodenum, jejunum, ileum, ascending colon, proximal 2/3 of the transverse colon  
Lower GI tract: distal 1/3 of the transverse colon, descending and sigmoid colon, rectum

## Viscerosomatic Reflexes Video



\*\*\*Another recommendation from past students/OMM Fellows.

## Distractors

**Answer A:** Counterstrain for a psoas tender point would not be the most appropriate choice for this postoperative patient. Positioning for counterstrain would likely increase his surgical pain. Because of this pain, it is highly unlikely that he would be able to remain passive and relaxed during this technique, which are requirements for successful counterstrain therapy.

**Answer B:** Mesentery myofascial release would likely benefit a patient with constipation, but it requires deep, intense manipulation of the bowels. Such manipulation would most likely not be tolerated and may even lead to complications in this patient with recent intra-abdominal surgery and incision.

**Answer C:** While muscle energy for the sacrum would theoretically help this patient, he would likely be unable to tolerate the positioning and muscle activation required for this therapy. Less aggravating options, such as paraspinal inhibition of the thoracolumbar junction, are more appropriate at this time.

**Answer D:** Myofascial release of the linea alba would most likely result in pulling on the surgical incision, either directly or indirectly. Although this treatment may improve freedom of motion of the intestines, it would likely not be tolerated by this patient, and it may even lead to surgical site complications.

## Question 13

56 y/o M presents to the office with a 3-week history of LBP. He recently started cycling on weekends but admits that longer rides are difficult because his bicycle seat is uncomfortable. He has also noticed more difficulty with urinating lately. PMH includes HTN and BPH. OSE reveals a tender point on the sacrum immediately superior and medial to the left ILA, with no tenderness along the midline of the sacrum. When documenting this finding, what is the most appropriate terminology for this tender point?

## Explanation and Bottom Line

Bottom Line: Posterior S5 tender points are located immediately superior and medial to the ILA on each side.

This patient has a **tender point** on the sacrum immediately superior and medial to the left inferior lateral angle (ILA), which corresponds to a left **posterior S5 tender point**. This finding is consistent with his difficulty with longer bike rides and his low back pain. His urinary symptoms are more likely related to benign prostatic hyperplasia.

For counterstrain treatment of a left posterior S5 tender point, posterior-to-anterior pressure would be applied at the opposite (right) sacral base, which would rotate the sacrum around the oblique axis. The position would then be fine-tuned until there is at least a 70% reduction in pain. This position is held for 90 seconds, followed by slow release of the pressure to allow the structures to return to their original position.

A tender point is thought to arise from overstretching a muscle or suddenly lengthening a shortened muscle, which leads to defensive contraction of the muscle to prevent injury. Microcirculation and tissue metabolism are affected in the area because of localized ischemia and the release of inflammatory cytokines, ultimately lowering the firing threshold of sensory neurons. This causes localized sensitization of the area, with edema and tenderness.

Locating a significant tender point is a necessary first step when considering counterstrain therapy. The below table summarizes the tender points located on the sacrum, with the numbers in parentheses corresponding to the numbers shown in the image below the table.

# Sacral Counterstrain

The below image shows the position of the sacral tender points.

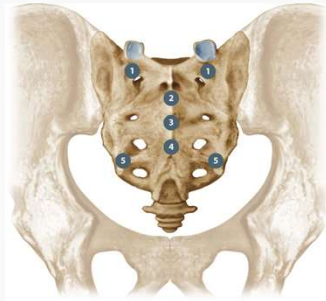


Illustration © TrueLearn, LLC

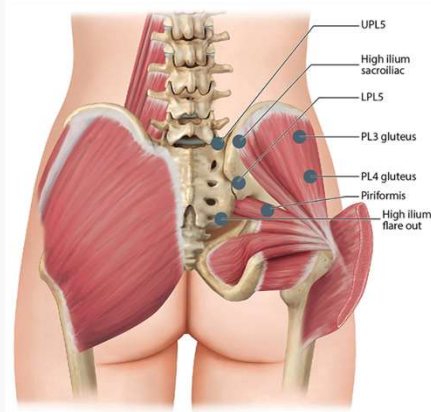
| Sacral Tender Points                                 |                                                                |
|------------------------------------------------------|----------------------------------------------------------------|
| Tender Point                                         | Location                                                       |
| Posterior S1 (1); 1 on each side                     | At each sacral sulcus                                          |
| Posterior S2 (2), posterior S3 (3), posterior S4 (4) | Midline on the sacrum at the corresponding level               |
| Posterior S5 (5); 1 on each side                     | Medial and superior to the inferior lateral angle on each side |

\*\*\*Pearl: THIS IS NOT TAUGHT AS PART OF TUCOM CURRICULUM so included it for you all so you can start reviewing it. Lots of overlap with Posterior lumbar and pelvis CS points so be able to differentiate all of these and know the anatomy. See next slide as well.

# Insight

## Insight

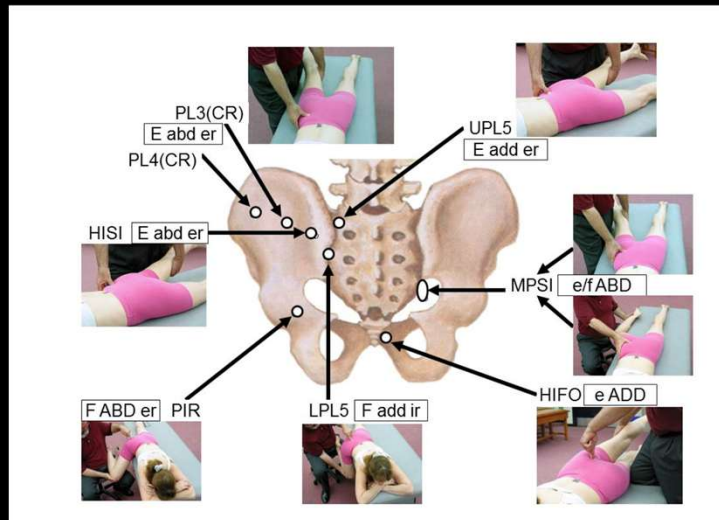
Be familiar with other similarly located tender points, as shown in the image below.



LPL5 = lower pole L5; PL3 = posterior L3; PL4 = posterior L4; UPL5 = upper pole L5.

| Counterstrain Tender Point Location, Test Vector & Tx Position |                                                                                          |               |             |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------|-------------|
| Posterior Pelvis                                               |                                                                                          |               |             |
| Counterstrain Points                                           | Counterstrain Point Location                                                             | Test Vector   | Tx Position |
| Upper Pole 5 <sup>th</sup> Lumbar (UPL5)                       | On the superior medial aspect of the PSIS                                                | SM → IL       | E add er    |
| High Ilium Sacroiliac (HISI)                                   | On the lateral aspect of the PSIS                                                        | L → M         | E abd er    |
| Posterior 3 <sup>rd</sup> Lumbar (PL3) - CR                    | Inferior to the iliac crest, half way between HISI & PL4                                 | P → A         | E abd er    |
| Posterior 4 <sup>th</sup> Lumbar (PL4) - CR                    | Inferior to the iliac crest, just posterior to the medial edge of the tensor fascia lata | P → A         | E abd er    |
| High Ilium Flareout (HIFO)                                     | On the lateral aspect of the coccyx                                                      | PL → AM (45°) | e ADD       |
| Midpole Sacroiliac (MPSI)                                      | Lateral to the ILA and inferior to the PSIS                                              | L → M         | e/f ABD     |
| Lower Pole 5 <sup>th</sup> Lumbar (LPL5)                       | On the inferior aspect of the PSIS                                                       | I → S         | F add ir    |
| Piriformis (PIR)                                               | In the belly of the muscle half way between sacrum & greater trochanter                  | P → A         | F ABD er    |

\*\*\*Pearl – they love the pelvis (both anterior and posterior) given their unique names that make them tricky to differentiate. Much more challenging on an exam than the C/T/L spine points in general.



## Distractors

**Answer A:** A high ilium flare out tender point is located at the lateral aspect of the ILA (as shown in the Insight section image). By contrast, this patient's tender point is located on the sacrum immediately superior and medial to the left ILA, corresponding to a left posterior S5 tender point.

**Answer B:** A high ilium sacroiliac tender point is located on the ilium, 2–3 cm lateral to the PSIS (as shown in the Insight section image). This patient's tender point is located on the sacrum instead.

**Answer C:** A lower pole L5 tender point is located on the ilium immediately inferior to the PSIS. This patient's findings are on the sacrum instead.

**Answer D:** A posterior S4 tender point is found in the midline on the sacrum at the level of S4. By contrast, the patient's findings are located on the left, adjacent to the ILA, and there is no tenderness along the midline.

## Question 14

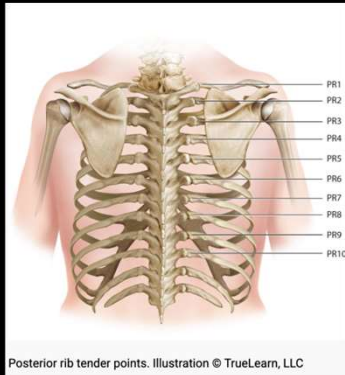
43 y/o F presents to the ED with left-sided chest pain. She reports that she had a cold last week but is otherwise healthy. She has no known cardiac risk factors, and initial cardiac workup is negative. She has pain with passive stretching of her left pec minor, and tenderness to palpation over the upper left anterolateral rib cage. There is a posterior rib 1 (PR1) tender point, which you treat with counterstrain. What position would you place the patient in for this treatment?

## Explanation and Bottom Line

Bottom Line: PR1 tender point is located on the posterosuperior aspect of rib 1 just lateral to the costotransverse articulation. Treatment of this involves the pt seated and the CT spine is used to induce sidebending away, rotation toward, and slight extension.

The patient likely has pectoralis strain with associated somatic dysfunctions in the ribs. A tender point is found on the posterior aspect of rib 1 (PR1). Counterstrain treatment of this involves the patient seated, and the cervical-thoracic spine is used to induce extension, side-bending away, and rotation toward. Just like any counterstrain treatment, the tender point is palpated and the pain level is assigned as a 10/10 to start. After adjusting into the position of ease, the tenderness is reassessed and small adjustments are made until the pain is reduced to at least a 3/10 (70% reduction). This position is held for 90 seconds, with gentle pressure on the tender point and without any active movement by the patient. The patient is positioned back into neutral, and the tender point is reassessed. The table below includes the locations and treatment positions of posterior rib tender points.

# Posterior Rib Counterstrain



| Location and treatment position of posterior rib (PR) tender points |                                                                         |                                                                |
|---------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------|
| Tender Point                                                        | Location                                                                | Treatment Position                                             |
| PR1                                                                 | Posterosuperior aspect of rib 1 lateral to costotransverse articulation | Patient seated; side-bend away, rotate toward, slightly extend |
| PR2-PR10                                                            | Posterosuperior angle of corresponding rib                              | Patient seated; flex, side-bend and rotate away                |

\*\*\*Please note that this goes lower than we typically teach you at TUCOM. We usually stop at PR6, but please note you can go to more inferior ribs, and the positioning is still the same.

## Distractors

**Answer A:** There is no counterstrain setup for anterior or posterior ribs in the prone position.

**Answer C:** This is the correct position for a posterior rib 2–10 counterstrain treatment.

**Answer D:** This is the correct setup for counterstrain of an anterior rib 3–10 tenderpoint.

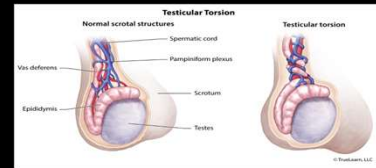
**Answer E:** This is the correct setup for counterstrain of an anterior rib 1 tender point.

## Question 15

14 y/o M brought to hospital at 3 am with sudden severe abdominal pain, nausea, vomiting, which woke him up from his sleep. He also complains of pain with urination. Vital signs below. PE reveals that the pt's left testicle is positioned higher than the right and is extremely tender to the touch. When gently stroking the right inner thigh, there is a subtle retraction of the right scrotum but no motion is visible on the left scrotum when the thigh is stroked. Which vertebral segments would correspond to the most likely viscerosomatic reflex in this condition?

|                |                  |
|----------------|------------------|
| Temperature    | 38.0°C (100.4°F) |
| Blood pressure | 133/78 mmHg      |
| Heart rate     | 115/min          |

# Explanation and Bottom Line



**Viscerosomatic reflexes** for the gonads are located in **T10-T11**.

**Testicular torsion** occurs when there is a twist within the spermatic cord which compromises blood flow and can cause testicular ischemia. It has been shown to happen in 3.8/100,000 males under the age of 18 years. It is a surgical emergency that may require orchiectomy. Patients frequently present with sudden testicular pain which may happen during a sports game or even in the middle of the night. Other common symptoms are fever, nausea, vomiting, abdominal pain, absent cremasteric reflex, or urinary symptoms.

When a patient presents with testicular torsion, a viscerosomatic reflex can be expected around T10-T11. However, a reflex in this area can also occur in various other conditions affecting the middle gastrointestinal (GI) tract, kidneys, and upper ureters; therefore, it is important for the provider to have a wide differential diagnosis.

**Bottom Line:** pt presenting with testicular torsion will often have a viscerosomatic reflex at T10-T11.

# Its back...

| Sympathetic Viscerosomatic Reflexes |                   |                 |                   |
|-------------------------------------|-------------------|-----------------|-------------------|
| Organ/Region                        | Spinal Cord Level | Organ/Region    | Spinal Cord Level |
| Head and neck                       | T1–T4             | Kidneys         | T10–T11           |
| Heart                               | T1–T5 (left)      | Upper ureters   | T10–T11           |
| Respiratory system                  | T2–T7             | Lower ureters   | T12–L1            |
| Esophagus                           | T2–T8             | Bladder         | T11–L2            |
| Upper GI tract                      | T5–T9             | Gonads          | T10–T11           |
| Middle GI tract                     | T10–T11           | Uterus/cervix   | T10–L2            |
| Lower GI tract                      | T12–L2            | Erectile tissue | T11–L2            |
| Appendix/cecum                      | T10–T12           | Prostate        | T12–L2            |
| Arms                                | T2–T8             | Legs            | T11–L2            |

Upper GI tract: stomach (left), liver and gallbladder (right), spleen (left), portions of the pancreas and duodenum  
Middle GI tract: portions of the pancreas and duodenum, jejunum, ileum, ascending colon, proximal 2/3 of the transverse colon  
Lower GI tract: distal 1/3 of the transverse colon, descending and sigmoid colon, rectum

\*\*\*You will see this chart pop up time and time again, so please know it!  
Autonomics of your ABCs!

## Distractors

**Answer A:** A viscerosomatic reflex found in T5-T9 often appears in the upper GI tract (eg, gastric ulcer, gastritis).

**Answer C:** A viscerosomatic reflex found in T10-T12 appears when there is a condition affecting the appendix or cecum (eg, appendicitis).

**Answer D:** A viscerosomatic reflex found in T11-L2 appears when there is a condition affecting the bladder (eg, interstitial cystitis, urinary tract infection).

**Answer E:** A viscerosomatic reflex found in T12-L2 appears when there is a condition affecting the lower GI tract (eg, ulcerative colitis flare).